

Instructions. (15 points) Sections 14.8, 15.1, 15.1b. Show relevant work for full credit.

- (4^{pts}) **1.** Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 2x + y$ subject to the constraint $x^2 + y^2 = 5$.

Solution:

Set $\nabla f = \lambda \nabla g$ where $g(x, y) = x^2 + y^2 - 5$:

$$2 = 2\lambda x \quad \text{and} \quad 1 = 2\lambda y$$

From the first equation: $\lambda = \frac{1}{x}$. Substituting into the second: $1 = \frac{2y}{x}$, so $x = 2y$.

Substitute into the constraint: $(2y)^2 + y^2 = 5 \Rightarrow 5y^2 = 5 \Rightarrow y = \pm 1$.

When $y = 1$: $x = 2$, $f(2, 1) = 5$.

When $y = -1$: $x = -2$, $f(-2, -1) = -5$.

Maximum = 5 at $(2, 1)$. Minimum = -5 at $(-2, -1)$.

- (4^{pts}) **2.** Approximate $\iint_R (x + 3y) dA$, where $R = [1, 3] \times [0, 6]$, by using a Riemann sum with a 2×2 grid of equal subrectangles and **midpoints** as sample points.

Solution:

$$\Delta x = \frac{3-1}{2} = 1, \quad \Delta y = \frac{6-0}{2} = 3, \quad \text{so } \Delta A = 1 \cdot 3 = 3.$$

The four midpoints are: $(1.5, 1.5)$, $(2.5, 1.5)$, $(1.5, 4.5)$, $(2.5, 4.5)$.

$$f(1.5, 1.5) = 1.5 + 4.5 = 6$$

$$f(2.5, 1.5) = 2.5 + 4.5 = 7$$

$$f(1.5, 4.5) = 1.5 + 13.5 = 15$$

$$f(2.5, 4.5) = 2.5 + 13.5 = 16$$

$$\iint_R (x + 3y) dA \approx (6 + 7 + 15 + 16)(3) = 44 \cdot 3 = \boxed{132}$$

- (3^{pts}) **3.** Without performing any integration, evaluate $\iint_R (3 - x) dA$ where $R = [0, 3] \times [0, 2]$.

Hint: Sketch the solid and identify its geometric shape. You may verify your answer using an iterated integral if you wish.

Solution:

The surface $z = 3 - x$ is a plane that slopes from $z = 3$ (at $x = 0$) down to $z = 0$ (at $x = 3$).

Over $R = [0, 3] \times [0, 2]$, this is a **triangular prism** (a wedge/ramp):

- Triangular cross-section has base 3 and height 3, so area = $\frac{1}{2}(3)(3) = \frac{9}{2}$.
- The prism extends 2 units in the y -direction.

$$\iint_R (3 - x) dA = \frac{9}{2} \cdot 2 = \boxed{9}$$

Verification: $\int_0^2 \int_0^3 (3 - x) dx dy = \int_0^2 \left[3x - \frac{x^2}{2} \right]_0^3 dy = \int_0^2 \frac{9}{2} dy = 9. \checkmark$

- (4^{pts}) **4.** Evaluate $\int x \cos x dx$.

Solution:

Integration by parts with $u = x$ and $dv = \cos x dx$:

$$du = dx, \quad v = \sin x.$$

$$\begin{aligned} \int x \cos x dx &= x \sin x - \int \sin x dx \\ &= x \sin x - (-\cos x) + C \\ &= \boxed{x \sin x + \cos x + C} \end{aligned}$$